



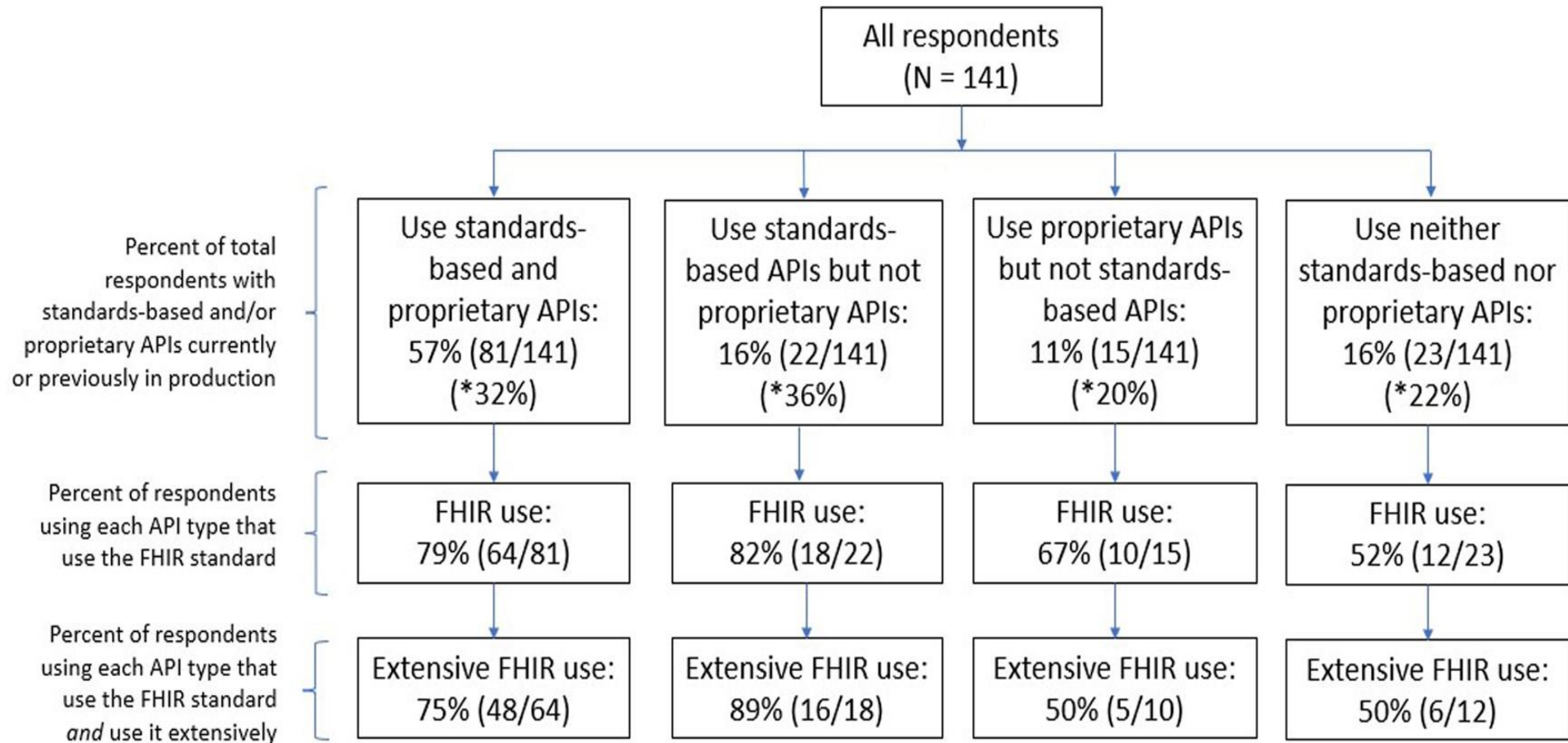
ELECTRONIC HEALTH RECORDS

FHIR: FAST HEALTHCARE INTEROPERABILITY RESOURCES

See also [FHIRAPI.ipynb](#) and [Abgabe1.ipynb](#)

75% of digital Health Companies are using FHIR

Digital health company use of APIs and the FHIR standard. N = 141.



- CDA has a very steep "learning curve"
- CDA is not very "developer friendly"
- HL7-v3 nearly not used for messages
 - Users stick to HL7-v2 messages
- HL7 standards do not have support for distributed information
 - You may have one blood pressure reading in Graz, another one in Innsbruck - how do you combine?

- Focus on implementation
 - *"From implementers for implementers"*
- Usual scenarios are kept simple
- Based on (for implementers) known technologies:
 - XML, JSON, RDF, RESTful web service
- Freely available to everyone
 - Was not always the case with HL7-v3 / CDA
- Demonstrates "best practices"



Background: History of FHIR

- HL7 v2 around since 1981 - very successful
- HL7 v3 about 10 years old, but apart from CDA has poor adoption
- FHIR grew out of frustration with v3
 - too hard for implementers (More for modellers)
 - too long to develop
 - inclusive
 - CDA good, but documents not enough
- Need a transition off v2
- Take all good ideas from v2/v3/CDA
- Mobile needs simple technology
- Fast Healthcare Interoperability Resources

<https://de.slideshare.net/HINZ/hay-introduction-to-hl7-fhir>

- Specification is written for a single target group: the implementers
- Reference implementations are immediately available
- Public test servers are available
- Starter APIs in different languages
(Java, C#, Delphi, ...) are published together with the specification
- Instance documents that are easy to understand
- Examples, examples, examples ...

- In FHIR, almost everything is a "resource"
 - Term from the semantic web world
- A basic unit, describing a "thing"
 - "Patient", "Prescription", "Device"
- FHIR tries to keep the number of "resources" limited
 - Differentiation through attributes / coding

- <https://www.hl7.org/fhir/resourcelist.html>

1.2 Resource Index

FHIR Infrastructure Work Group	Maturity Level: N/A	Standards Status: Informative
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This page is provided to help find resources quickly. There is also a more [detailed classification, ontology, and description](#). For background to the layout on the layers in this page, see the [Architect's Overview](#). See also the abstract Base Resources [Resource](#) and [DomainResource](#).

Categorized							Alphabetical	R2 Layout	By Maturity	Security Category	By Standards Status	By Committee
Foundation	Conformance		Terminology		Security		Documents		Other			
	- CapabilityStatement N - StructureDefinition N - ImplementationGuide 1 - SearchParameter 3 - MessageDefinition 1 - OperationDefinition N - CompartmentDefinition 1 - StructureMap 2 - GraphDefinition 1 - ExampleScenario 0		- CodeSystem N - ValueSet N - ConceptMap 3 - NamingSystem 1 - TerminologyCapabilities 0		- Provenance 3 - AuditEvent 3 - Consent 2		- Composition 2 - DocumentManifest 2 - DocumentReference 3 - CatalogEntry 0		- Basic 1 - Binary N - Bundle N - Linkage 0 - MessageHeader 4 - OperationOutcome N - Parameters N - Subscription 3			
	Individuals		Entities #1		Entities #2		Workflow		Management			
	- Patient N - Practitioner 3		- Organization 3 - OrganizationAffiliation 0		- Substance 2 - BiologicallyDerivedProduct 0		- Task 2 - Appointment 3		- Encounter 2 - EpisodeOfCare 2			

- Is there a FHIR resource for "disease"?
- If not, how is the FHIR resource for "disease" named?
 - Why isn't it named "disease"?
- Navigate through the resource description and try to understand how the documentation works

- What does the "N" on a resource mean?
- What does the number on a resource mean?
- Which are the resources meant for use in clinical research?
- How does one define a specific (e.g.) lab test (not taken yet, no patient yet)

- For patients that enroll in a clinical research study, what are the resources used?
- Tips:
 - start from "ResearchStudy" and "ResearchSubject"
 - look for references and "referenced by"
- Make a diagram that contains all these resources and their relations

- “FHIR Resources can be thought of as the "atoms" of health information exchange.”
- Smallest unit of exchange
- Resource definition:
<https://www.hl7.org/fhir/resource.html>

Categorized	Alphabetical	R2 Layout	By Maturity	Security Category	By Standards Status	By Work Group
Level 5						
• Binary N			• DomainResource N	• OperationOutcome N		• QuestionnaireResponse
• Bundle N			• Immunization	• Organization		• RelatedPerson
• CanonicalResource N			• Location	• Parameters N		• Resource N
• CapabilityStatement N			• MetadataResource N	• Patient N		• SearchParameter
• CodeSystem N			• Observation N	• Practitioner		• StructureDefinition N
• Condition			• OperationDefinition N	• Questionnaire		• ValueSet N
Level 4						
• ActivityDefinition			• HealthcareService	• MedicationRequest		• PractitionerRole
• AuditEvent			• ImagingStudy	• MedicationStatement		• Procedure
• Composition			• ImplementationGuide	• MessageHeader		• Provenance
• Coverage			• Library	• NamingSystem		• RequestOrchestration
• CoverageEligibilityRequest			• List	• PaymentNotice		• ServiceRequest
• CoverageEligibilityResponse			• Measure	• PaymentReconciliation		• StructureMap
• DocumentReference			• MeasureReport	• Person		• TestScript
• Encounter			• Medication	• PlanDefinition		
Level 3						
• AllergyIntolerance			• CompartmentDefinition	• MedicinalProductDefinition		• Task
• Appointment			• ConceptMap	• Schedule		• VisionPrescription
• AppointmentResponse			• DiagnosticReport	• Slot		
• Basic			• Group	• Subscription		

- Scope and usage:
<https://www.hl7.org/fhir/patient.html#scope>

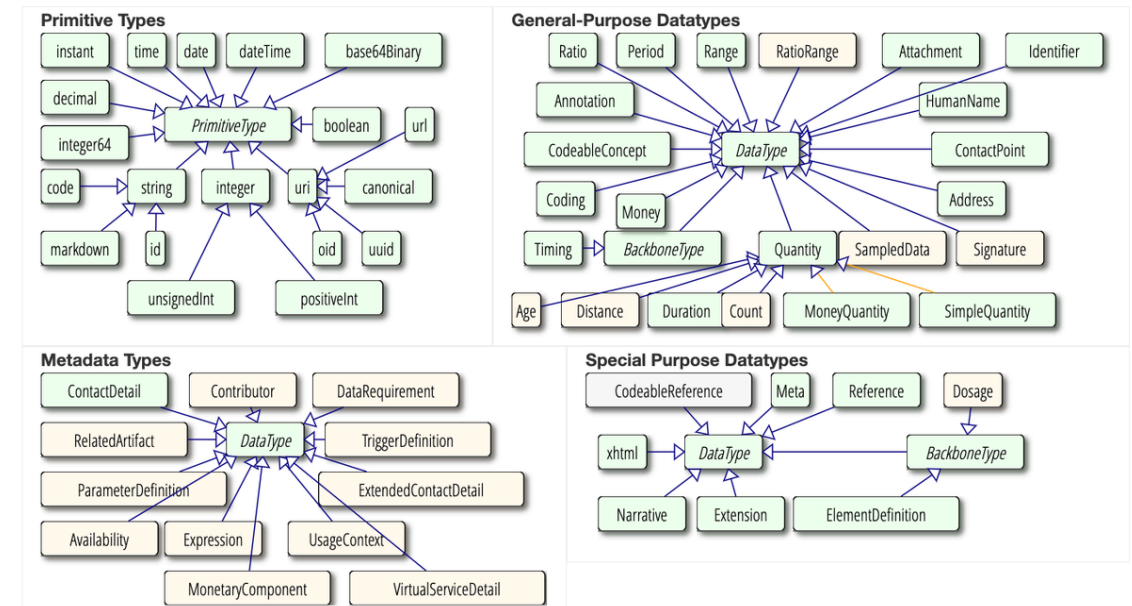
Structure	UML	XML	JSON	Turtle	R4 Diff	All
Structure						
Name	Flags	Card.	Type	Description & Constraints		
Patient	N		DomainResource	Information about an individual or animal receiving health care		
identifier	Σ	0..*	Identifier	Elements defined in Ancestors: id , meta , implicitRules , An identifier for this patient		
active	?! Σ	0..1	boolean	Whether this patient's record is in active use		
name	Σ	0..*	HumanName	A name associated with the patient		
telecom	Σ	0..*	ContactPoint	A contact detail for the individual		
gender	Σ	0..1	code	male female other unknown Binding: AdministrativeGender (Required)		
birthDate	Σ	0..1	date	The date of birth for the individual		
deceased[x]	?! Σ	0..1		Indicates if the individual is deceased or not		
deceasedBoolean			boolean			
deceasedDateTime			dateTime			
address	Σ	0..*	Address	An address for the individual		

- Data Elements that define what data points are to be captured as part of the Patient resource (identifier, active, and name)
- <https://www.hl7.org/fhir/patient-definitions.html>
- Element Id tells you what this data point will be called in the final resource
- Definition, Short Display, and Note explain how the field is supposed to be used

Each data element has a **Type**. For example, the identifier is of the [Identifier](#) type, and clicking on the link takes you to the definition of what this type means

Name	Flags	Card.	Type
 Patient	N		DomainResource
 identifier	Σ	0..*	Identifier
 active	?! Σ	0..1	boolean

- Some types are simple (or primitive) like active, which is a boolean (true or false), while others like identifier and name have another nested structure within and are complex



Structure	UML	XML	JSON	Turtle	R4 Diff	All
Structure						
Name	Flags	Card.	Type	Description & Constraints		
 Identifier	Σ N		Element	An identifier intended for computation + <i>Warning: Identifier with no value has limited utility. If the value element SHOULD be present with an extension</i>		
 use	?! Σ	0..1	code	Elements defined in Ancestors: id , extension usual official temp secondary old (If known) Binding: IdentifierUse (Required)		
 type	Σ	0..1	CodeableConcept	Description of identifier Binding: Identifier Type Codes (Extensible)		
 system	Σ	0..1	uri	The namespace for the identifier value		
 value	Σ C	0..1	string	The value that is unique		
 period	Σ	0..1	Period	Time period when id is/was valid for use		
 assigner	Σ	0..1	Reference(Organization)	Organization that issued id (may be just text)		

Contains other complex types like CodeableConcept and Period

<https://www.hl7.org/fhir/datatypes.html>

Primitive Types

FHIR Name Value Domain

base64Binary A stream of bytes, base64 encoded ([RFC 4648](#)). base64Binary content does not include any whitespace or line feeds, but reading applications should ignore whitespace characters (per [RFC 4648](#))
Regex: `(?:[A-Za-z0-9+/]{4})*(?:[A-Za-z0-9+/]{2}==| [A-Za-z0-9+/]{3}=)?`
There is no specified upper limit to the size of a binary, but systems will have to impose some implementation based limit to the size they support. This should be clearly documented, though there is no computable way for this at this time

boolean true | false

Regex: `true|false`

canonical A URI that refers to a [resource by its canonical URL](#) (resources with a [url](#) property). The `canonical` type differs from a `uri` in that it has special meaning in this specification, and in that it may have a version appended, separated by a vertical bar (|). Note that the type `canonical` is not used for the actual canonical URLs that are the target of these references, but for the URIs that refer to them, and may have the version suffix in them. Like other URIs, elements of type `canonical` may also have #fragment references. Unlike other URIs, canonical URLs are never relative - they are either absolute URIs, or fragment identifiers
Regex: `\S*`

code Indicates that the value is taken from a set of controlled strings defined elsewhere (see [Using codes](#) for further discussion). Technically, a code is restricted to a string which has at least one character and no leading or trailing whitespace, and where there is no whitespace other than single spaces in the contents
Regex: `^[^s]+C [^s]+)*`
This datatype can be [bound](#) to a [ValueSet](#)

date A date, or partial date (e.g. just year or year + month) as used in human communication. The format is a subset of [\[ISO8601\]](#): YYYY, YYYY-MM, or YYYY-MM-DD, e.g. 2018, 1973-06, or 1905-08-23. **There SHALL be no timezone offset.** Dates SHALL be valid dates.

Regex:
`([0-9]([0-9]([0-9][1-9]| [1-9]0)| [1-9]00)| [1-9]000)(-([0-9]| 1[0-2])(-([0-9]| 1[2][0-9]| 3[0-1])?)?)?`

- The "*Card.*" field stands for **Cardinality** and represents the minimum and maximum times a particular *data element* can appear using the notation minimum..maximum. A maximum value of * means "as many times as needed" or ∞ .

Name	Flags	Card.	Type
 Patient			DomainResource
 identifier	Σ	0..*	Identifier
 active	?! Σ	0..1	boolean

```
{  
  "resourceType": "Patient"  
}
```

The resourceType field always shows up in any resource, and it's always the name of the FHIR resource. So a FHIR Patient resource will implicitly need to have a resourceType of "Patient" at all times.

"base" version of the FHIR standard extendable to as many use cases as possible

- international FHIR specification need to be constrained further for local use—usually within a nation, an organization, or
- implementers to constrain and extend the base specification of a resource for a particular use case

- <https://build.fhir.org/ig/HL7/US-Core/StructureDefinition-us-core-patient.html>
- <https://www.healthit.gov/test-method/standardized-api-patient-and-population-services>

13.148.1.2 Formal Views of Profile Content

Description of Profiles, Differentials, Snapshots and how the different presentations work [↗](#).

Differential Table	Key Elements Table	Snapshot Table	Statistics/References	All
This structure is derived from Patient ↗				
Name	Flags	Card.	Type	Description & Constraints
Patient		0..*	Patient	Information about an individual or animal receiving health care services
Slices for extension				Content/Rules for all slices
race		0..1	(Complex)	ADDITIONAL USCDI: US Core Race Extension. (multiple races are supported in the e URL: http://hl7.org/fhir/us/core/StructureDefinition/us-core-race
ethnicity		0..1	(Complex)	ADDITIONAL USCDI: US Core ethnicity Extension (multiple ethnicities are supported URL: http://hl7.org/fhir/us/core/StructureDefinition/us-core-ethnicity
tribalAffiliation		0..*	(Complex)	ADDITIONAL USCDI: Tribal Affiliation Extension URL: http://hl7.org/fhir/us/core/StructureDefinition/us-core-tribal-affiliation
birthsex		0..1	code	Birth Sex Extension URL: http://hl7.org/fhir/us/core/StructureDefinition/us-core-birthsex Binding: Birth Sex ↗ (required): Code for sex assigned at birth
sex		0..1	code	ADDITIONAL USCDI: Sex Extension URL: http://hl7.org/fhir/us/core/StructureDefinition/us-core-sex Binding: Sex ↗ (extensible): Concepts used for general documentation of sex rep
genderIdentity		0..*	CodeableConcept	ADDITIONAL USCDI: The individual's gender identity URL: http://hl7.org/fhir/us/core/StructureDefinition/us-core-genderIdentity Binding: Gender Identity ↗ (extensible)
identifier	S	1..*	Identifier	An identifier for this patient
system	S	1..1	uri	The namespace for the identifier value
value	S	1..1	string	The value that is unique within the system.
name	S C	1..*	HumanName	A name associated with the patient us-core-6: At least name.given and/or name.family are present or, if neither is av is present.
use		0..1	code	ADDITIONAL USCDI: usual official temp nickname anonymous old maiden
family	S C	0..1	string	Family name (often called 'Surname')
given	S C	0..*	string	Given names (not always 'first'). Includes middle names
suffix		0..*	string	ADDITIONAL USCDI: Parts that come after the name
period		0..1	Period	ADDITIONAL USCDI: Time period when name was/is in use

- There are extra fields like race and ethnicity that were not present in the base FHIR resource.
- The cardinality of the identifier is now 1..*. The identifier element for a patient is not optional anymore. It is **mandatory**.
- You'll notice the "S" square in red. This stands for "must support," and it basically means systems should be ready to handle this information.

What is the difference between Must Support and Mandatory?

- Extensions are a way to add new elements to a resource without changing its core definition.
 - They allow for flexibility and customization of FHIR resources to meet specific needs or requirements that aren't covered by the standard elements.
-
1. They have a defined URL that identifies the extension.
 2. They can contain any datatype allowed in FHIR.
 3. They can be simple or complex, containing multiple sub-extensions.

```
{
  "resourceType": "Patient",
  "extension": [
    {
      "url": "http://hl7.org/fhir/us/core/StructureDefinition/us-core-race",
      "extension": [
        {
          "url": "ombCategory",
          "valueCoding": {
            "system": "urn:oid:2.16.840.1.113883.6.238",
            "code": "2054-5",
            "display": "Black or African American"
          }
        },
        {
          "url": "text",
          "valueString": "Black or African American"
        }
      ]
    }
  ],
  // ... other Patient elements ...
}
```

- Do not try to describe the whole medical world ...
 - But only 80% of it ...
- Do not try to describe concepts that are cultural dependent
 - Race
 - Religion
 - ...
- The 20 other % can be described using **extensions**
 - Can be local in scope

- US census bureau:
 - White
 - Black or african America
 - American Indian and Alaska Native
 - Asian
 - Native Hawaiian and Other Pacific Islander – Hispanic or Latino

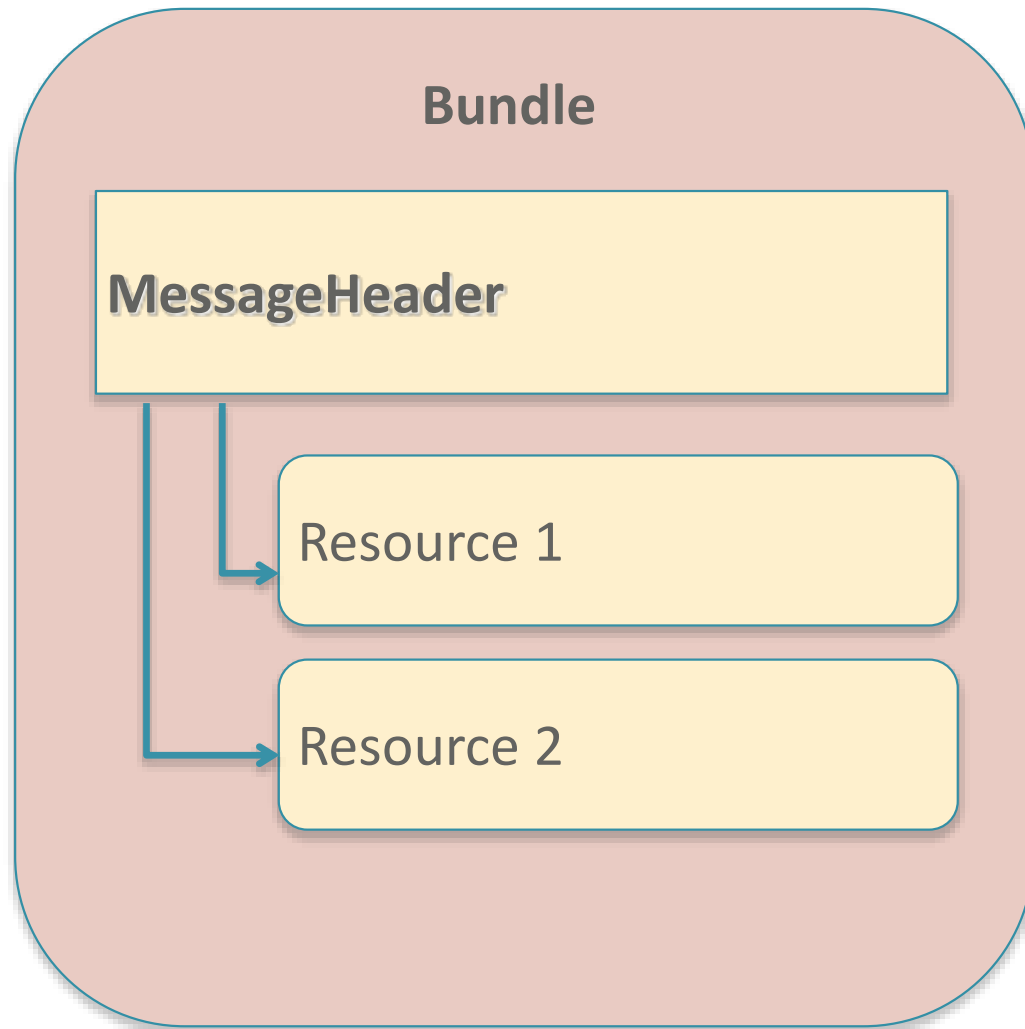
- Important: "self-reported"

- Foundation
 - ValueSet, Composition, MessageHeader, ...
- Base
 - Patient, Practitioner, Encounter, ...
- Clinical
 - Condition, Medication, Specimen, ...
 - Financial
 - Claim, Contract, ...
- Specialized
 - ResearchSubject ...

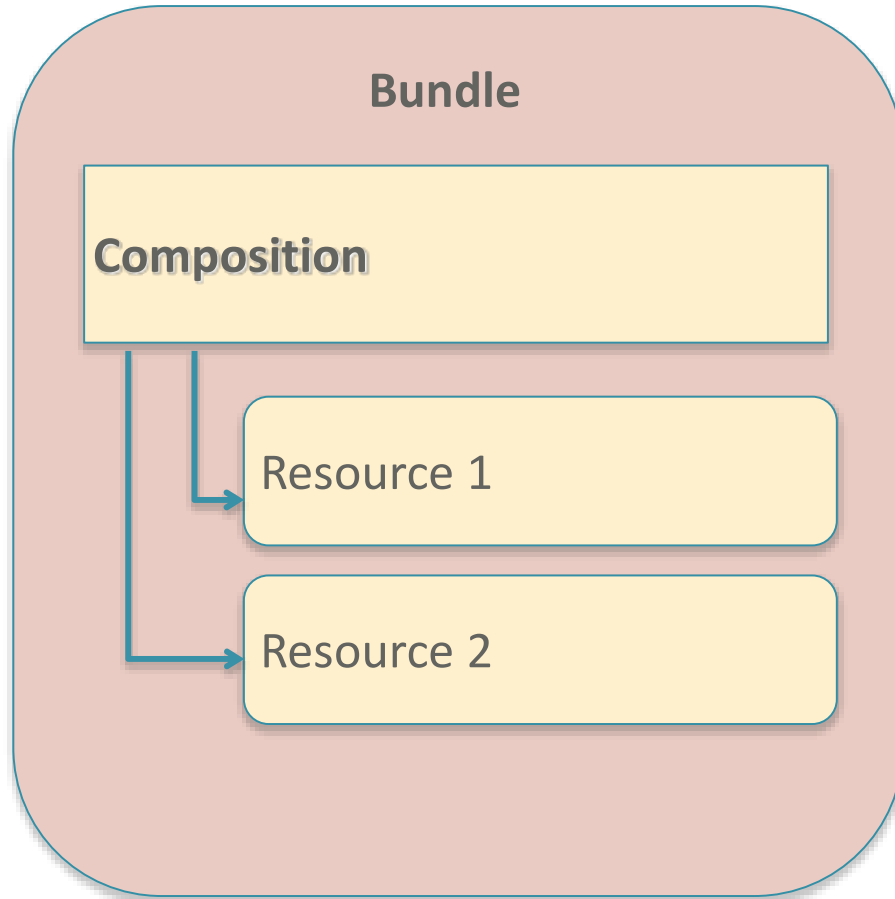
- FHIR wants to keep the number of resources limited
- Some resources cover a broad scope
- Example: "Observation" resource
 - "S" and "O" in SOAP ...
- Differentiation by attributes and coding

- How does FHIR code "weight" in an "Observation Resource"?
- Are there other coding systems used?

- Documents and messages are equivalent
- First resource in "Bundle" defines whether the intention is a "document" or a "message"
- Simple exchange of first resource within a Bundle transforms a message into a document and vice versa (see later)

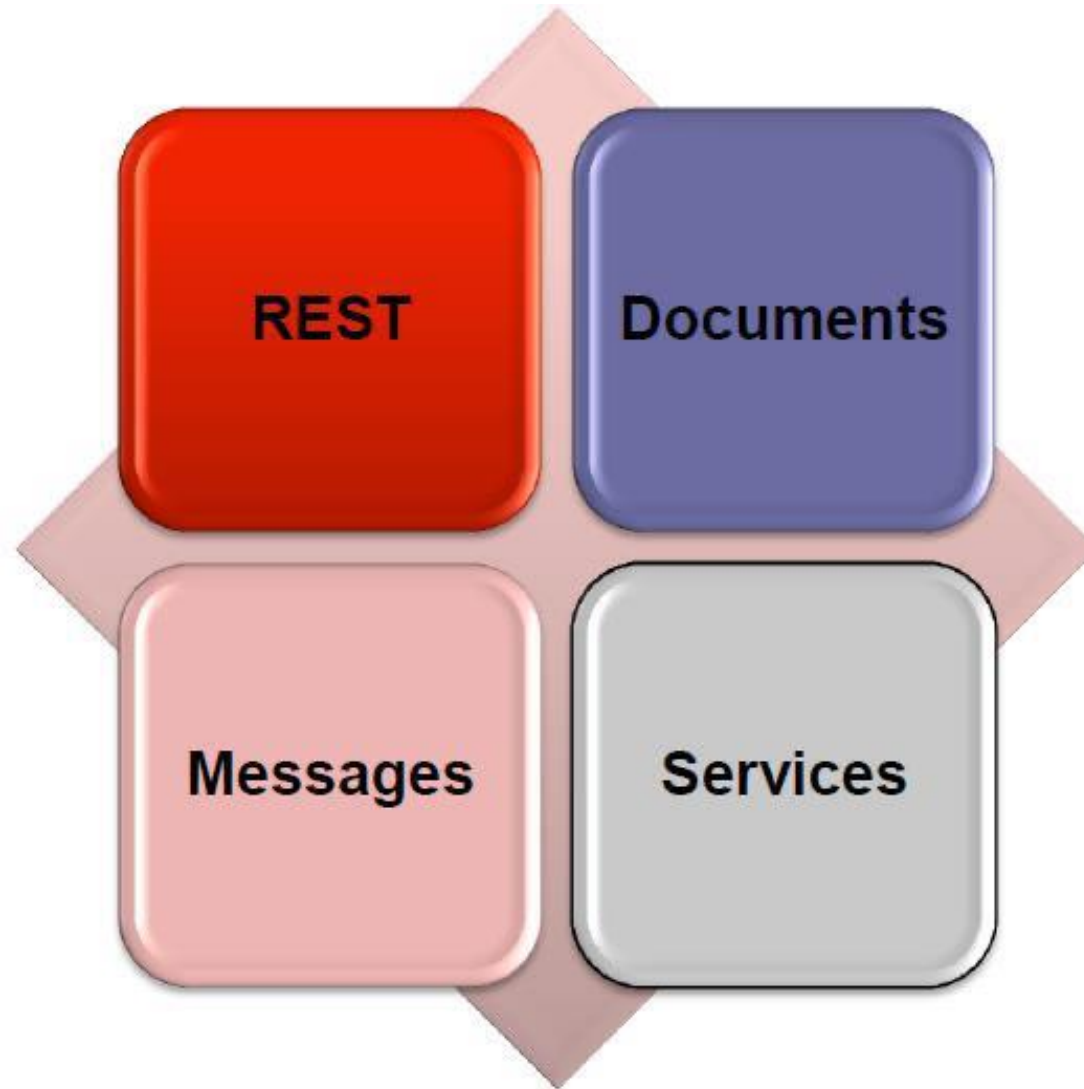


- Similar to v2 and v3 messaging
- Collection of resources bound together
 - Root is a “MessageHeader” resource
- Allows request/response behavior with bundles for both request and response
- Event-driven
 - E.g. Send lab order, get back result
- Can use HTTP
- Can be asynchronous



- Similar to CDA
- Also a collection of resources as a Bundle
 - Root is a “Composition” resource
 - Just like CDA header
- Explicit references
- Can be signed, authenticated, etc.

- Messages and documents are equivalent
- Medical information can be located anywhere
- You can have a BP reading in Graz and another one in Patagonia
- Use modern (machine-machine) communication technologies
 - RESTful web services
- Support modern data formats: JSON, XML, RDF-Turtle



- In CDA, there is little support for referencing other "objects"
- The same medical doctor can be described several times within a single CDA document
 - Can be in different participations

- In FHIR, a "thing" (Resource) is defined/described once and can be referenced many times
- **Exercise:** take an "Observation" example, and find out how the subject of the observation (the patient) is referenced

- References are always URLs
 - Absolute or relative
 - Internal or external
- External reference - absolute URL
 - <https://..../Patient?>...
- Internal reference - relative URL
 - Patient/example

See: <https://www.hl7.org/fhir/references.html>

- A relative reference to the [Patient](#) "034AB16" in an element named subject on a FHIR RESTful server:

```
<subject>
  <reference value="Patient/034AB16" />
</subject>
```

- An absolute reference to a [Structure Definition](#) in an element named profile:

```
{
  "profile" : {
    "reference" : "http://fhir.hl7.org/svc/StructureDefinition/c8973a22-2b5b-4e76-9c66-00639c99e61b"
  }
}
```


- References can be version specific - that is, a reference may point to a specific version of a resource. e.g.:

```
<target>  
  <reference value="http://example.org/fhir/Observation/1x2/_history/2" />  
</target>
```

- External URL (absolute) must not be a FHIR RESTFul server
- However: using a RESTful web server is the preferred approach
- References MUST always point to a Resource

- `https://.../<resourcename>/<identifier>`

- **Examples:**

`<profile>`

`<reference value="http://fhir.hl7.org/svc/StructureDefinition/c8 973a22-2b5b-4e76-9c66-00639c99e61b" />`

`</profile>`

- REST = Representational state transfer
- Stateless operations
 - each request is handled as an independent transaction (no remembrance of state)
- Request/response
 - Requests are made to a resource's URL
 - Response comes as XML, HTML, JSON or other predefined format
- GET, POST, PUT, DELETE
 - So resources can also be altered/deleted on the target server ...

- For use with relative references
 - No obligation, but recommended
- Allow to retrieve, search, update, delete ... individual resources
- Several reference implementations available

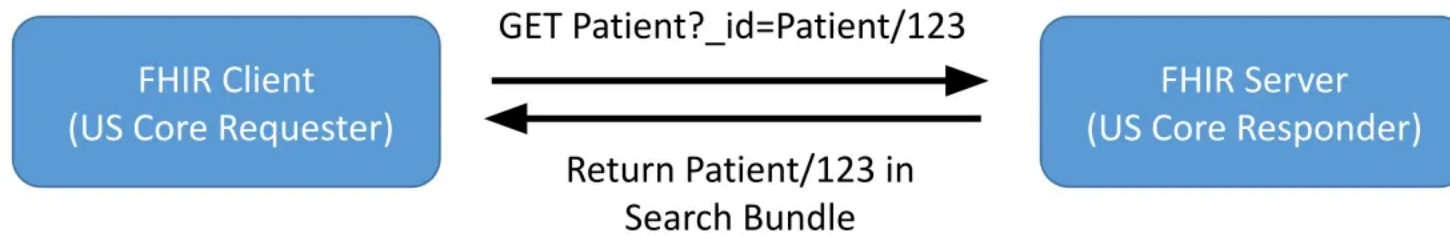
<https://www.hl7.org/fhir/http.html>

- Which operations are available?
- How is security handled?
- What return formats are supported?
- What is returned when a resource is not found?

<https://www.hl7.org/fhir/http.html>

- resources should be exchanged and consumed by different systems
- **RESTful, Documents, and Messaging**

- standardized way to interact with FHIR resources over HTTP
- Key aspects of FHIR RESTful APIs include:
 - CRUD operations (Create, Read, Update, Delete) on resources
 - Search functionality with standardized parameters
 - Versioning of resources
 - Batch and transaction support for multiple operations
- All CRUD operations on a FHIR server can be done using similar HTTP operations



- Primary mechanism:
GET [base]/{resourcename}?a=b

- Where:
 - [base] is the base URI, e.g.
 - <https://myfhirserver.com>
 - {resourcename} is the name of the resource
 - ? is the "where" statement
 - a=b is the query pair

- Example:
 - https://myfhirserver.com/Patient?_id=1234

- `_content`
- `_id`
- `_lastUpdated`
- `_profile`
- `_query`
- `_security`
- `_tag`
- `_text`
- `_filter`

2.28.4 Search Parameters

Common search parameters defined by this resource. See [Searching](#) for more information about searching in REST, messaging, and services.

Name	Type	Description	Paths
_content	string	Search on the entire content of the resource	
_id	token	Logical id of this artifact	Resource.id
_lastUpdated	date	When the resource version last changed	Resource.meta.lastUpdated
_profile	uri	Profiles this resource claims to conform to	Resource.meta.profile
_query	token	A custom search profile that describes a specific defined query operation	
_security	token	Security Labels applied to this resource	Resource.meta.security
_tag	token	Tags applied to this resource	Resource.meta.tag

- Search behaviour depends on the data type
 - String → part of ...
so not looking for the exact match
 - /Patient?name=Williams
looks for all resources for which the patient name (in one way or another) contains "Williams" (case insensitive)

2.21.1.4.3 Search Parameter Types

Each search parameter is defined by a type that specifies how the search parameter behaves. These are the defined parameter types:

number	Search parameter SHALL be a number (a whole number, or a decimal).
date	Search parameter is on a date/time. The date format is the standard XML format, though other formats may be supported.
string	Search parameter is a simple string, like a name part. Search is case-insensitive and accent-insensitive. May match just the start of a string. String parameters may contain spaces.
token	Search parameter on a coded element or identifier. May be used to search through the text, displayname, code and code/codesystem (for codes) and label, system and key (for identifier). Its value is either a string or a pair of namespace and value, separated by a " ", depending on the modifier used.
reference	A reference to another resource.
composite	A composite search parameter that combines a search on two values together.
quantity	A search parameter that searches on a quantity.
uri	A search parameter that searches on a URI (RFC 3986).

<https://www.hl7.org/fhir/search.html>

- Resource?parameter:modifier=term
- Patient?give:exact=John
- Patient?given:contains=ilia
- Patient?gender:missing=true

- For all parameters (except combination): `:missing` ; e.g. `gender:missing=true` (or false). Searching for `gender:missing=true` will return all the resources that don't have a value for the gender parameter (which usually equates to not having the relevant element in the resource). Searching for `gender:missing=false` will return all the resources that have a value for the `gender` parameter. For simple data type elements, `:missing=true` will match on all elements where either the underlying element is omitted or where the element is present with extensions but no @value is specified
- For string: `:exact` returns results that match the entire supplied parameter, including casing and combining characters. Note that the handling of (extended) [grapheme clusters](#) is within the discretion of the server, i.e. the server decides if a string search parameter matches on canonically equivalent characters, or matches on the actual used Unicode code points, or `:contains` (case insensitive and combining character-insensitive, search text matched anywhere in the string), instead of the default behavior (case insensitive and accent-insensitive, partial matches at the start of the string).
- For token: `:text` (the match does a partial searches on the text portion of a CodeableConcept or the display portion of a Coding), instead of the default search which uses codes. Other defined modifiers are `:in`, `:below`, `:above` and `:not-in` which are described below.
- For reference: `:[type]` where [type] is the name of a type of resource, `:identifier`, and, for some parameters, `:above` and `:below`
- For uri: `:below`, `:above` indicate that instead of an exact match, either the search term left-matches the value, or vice-versa.

Server SHALL reject any search request that contains is suffixed by a modifier that the server does **not** support for that parameter. For example, if the server supports the `name` search parameter, but not the `:exact` modifier on the name, it should reject a search with the parameter `name:exact=Bill`, using an HTTP `400` error with an [OperationOutcome](#) with a [clear error message](#).

- When successful, the found resources are always returned as a "Bundle"
 - Either as XML, JSON or RDF
- Individual resources have additional search parameters

- You can use all fields from the "Patient" resource
<https://www.hl7.org/fhir/patient-definitions.html>

Patient.name	
Definition	A name associated with the individual.
Control	0..*
Type	HumanName
Requirements	Need to be able to track the patient by multiple names. Examples are your official name and a partner name.
Summary	true
Comments	A patient may have multiple names with different uses or applicable periods. For animals, the name is a "HumanName" in same patterns.

Patient.telecom	
Definition	A contact detail (e.g. a telephone number or an email address) by which the individual may be contacted.
Control	0..*
Type	ContactPoint
Requirements	People have (primary) ways to contact them in some way such as phone, email.
Summary	true
Comments	A Patient may have multiple ways to be contacted with different uses or applicable periods. May need to have options for identification. The address may not go directly to the individual, but may reach another party that is able to proxy for the

Patient.gender	
Definition	Administrative Gender - the gender that the patient is considered to have for administration and record keeping purpose
Control	0..1
Terminology Binding	AdministrativeGender (Required)
Type	code



You are accessing the public FHIR server **UHN_HAPI Server (R4 FHIR)**. This server is hosted elsewhere on the internet but is being accessed using the HAPI client implementation.

⚠ This is not a production server! Do not store any information here that contains personal health information or any other confidential information. This server will be regularly purged and reloaded with fixed test data.

Server	UHN Test Server (R4 Resources)
Software	HAPI FHIR Server - 5.0.0-SNAPSHOT
FHIR Base	http://hapi.fhir.org/baseR4

- Base: <http://hapi.fhir.org/baseR4>
- Look for all patients with the name "Andrew"
- Look for the patient with ID "697701"
- Look for all the "male"/"female" patients
- Look for all patients born before 1950

- Base: <http://hapi.fhir.org/baseR4>
- Look for all patients with the first name ("given name") "Sally"
- search for all female patients with the name "Andrew" that are "female"
- Look for all patients with the family name "Johnson"

- Base: <http://hapi.fhir.org/baseDstu3>
- Look for the allergies of patient 1707602
- What substance is the patient allergic to?
- What category is that?

- You can choose the format of the returned resources using the parameter "_format"
 - _format=xml
 - _format=json
 - _format=rdf (not always supported)
- Exercise
 - Repeat the previous exercise, but now ask for XML to be returned

- Base: <http://hapi.fhir.org/baseDstu3>
- Look for all allergies of patient 1707602 of which the status is active
- Try to find patients that have allergies that have the status "inactive"

- Base: <http://hapi.fhir.org/baseDstu3>
- Find the allergies of patients with the id "1707602" (this is the ID of the patient, not of the allergy)
- Try to find all allergies of the patient with the name "Andrew"

9.1.4 Search Parameters

Search parameters for this resource. The [common parameters](#) also apply. See [Searching](#) for more information about searching in REST, messaging, and services.

Name	Type	Description	Expression	In Common
asserter	reference	Source of the information about the allergy	AllergyIntolerance.asserter (Practitioner , Patient , PractitionerRole , RelatedPerson)	
category	token	food medication environment biologic	AllergyIntolerance.category	
clinical-status	token	active inactive resolved	AllergyIntolerance.clinicalStatus	
code	token	Code that identifies the allergy or intolerance	AllergyIntolerance.code AllergyIntolerance.reaction.substance	13 Resources
criticality	token	low high unable-to-assess	AllergyIntolerance.criticality	
date	date	Date first version of the resource instance was recorded	AllergyIntolerance.recordedDate	17 Resources
identifier	token	External ids for this item	AllergyIntolerance.identifier	30 Resources
last-date	date	Date(/time) of last known occurrence of a reaction	AllergyIntolerance.lastOccurrence	
manifestation	token	Clinical symptoms/signs associated with the Event	AllergyIntolerance.reaction.manifestation	
onset	date	Date(/time) when manifestations showed	AllergyIntolerance.reaction.onset	
patient	reference	Who the sensitivity is for	AllergyIntolerance.patient	33 Resources

<https://www.hl7.org/fhir/allergyintolerance.html>

- FHIR has a "general" set of search parameters
 - Available for ALL resources, e.g. "_id"

<https://www.hl7.org/fhir/search.html>

3.1.1.1 Summary Table

Search Parameter Types	Parameters for all resources	Search result parameters
Number	_id	_sort
Date/DateTime	_lastUpdated	_count
String	_tag	_include
Token	_profile	_revinclude
Reference	_security	_summary
Composite	_text	_total
Quantity	_content	_elements
URI	_list	_contained
Special	_has	_containedType
	_type	

- Each resource has a number of resource-specific search parameters
 - Described in the resource description itself

Exercise: what are the search parameters for "AllergyIntolerance"?

For the ordered parameter types of **number**, **date**, and **quantity**, a prefix to the parameter value may be used. To avoid confusion, the following prefixes are used:

eq	the value for the parameter in the resource is equal to the provided value	the range the target value
ne	the value for the parameter in the resource is not equal to the provided value	the range the target value
gt	the value for the parameter in the resource is greater than the provided value	the range the target value
lt	the value for the parameter in the resource is less than the provided value	the range the target value
ge	the value for the parameter in the resource is greater or equal to the provided value	the range the target value the target value
le	the value for the parameter in the resource is less or equal to the provided value	the range the target value the target value
sa	the value for the parameter in the resource starts after the provided value	the range value, a value
eb	the value for the parameter in the resource ends before the provided value	the range value, a value
ap	the value for the parameter in the resource is approximately the same to the provided value. Note that the recommended value for the approximation is 10% of the stated value (or for a date, 10% of the gap between now and the date), but systems may choose other values where appropriate	the range

- [base]/Observation?...&value-quantity=**gt**100
- [base]/Encounter?date=**le**2017-04-02

- FHIR Document Bundles allow for the exchange of collections of resources as a single unit
- *“FHIR Bundles are a way to package multiple resources into a single larger resource.”*
- Contain a Composition resource (which is another resource) as the first entry, providing context
- Include all referenced resources within the bundle
- Are immutable once created
- Can be digitally signed for integrity and non-repudiation

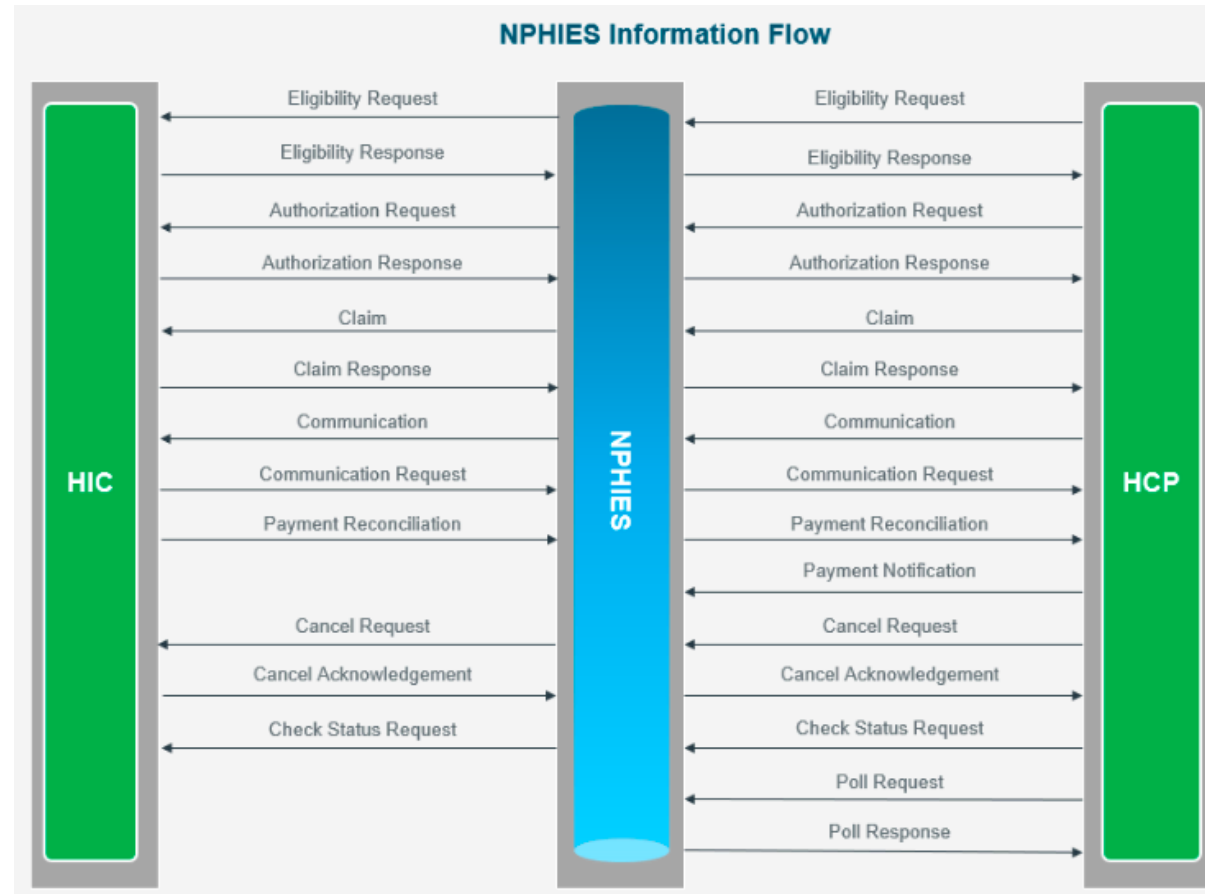
```
{
  "resourceType": "Bundle",
  "type": "document",
  "entry": [
    {
      "resource": {
        "resourceType": "Composition",
        // ... Composition details ...
      }
    },
    // ... Other resources ...
  ]
}
```

- <https://nrces.in/ndhm/fhir/r4/StructureDefinition-DischargeSummaryRecord.html>

- enables event-based communication between systems
- where actions need to be triggered based on certain events or where asynchronous communication is preferred
 - FHIR Messaging uses Bundles just like FHIR Documents. Where they differ is that FHIR Messaging:
 - Uses a MessageHeader resource to define the purpose and content of the message
 - Can include multiple resources related to the event
 - Supports request/response patterns
 - Allows for routing and reliable delivery mechanisms

```
{
  "resourceType": "Bundle",
  "type": "message",
  "entry": [
    {
      "resource": {
        "resourceType": "MessageHeader",
        // ... MessageHeader details ...
      }
    },
    // ... Other resources ...
  ]
}
```

FHIR Messaging paradigm in practice would be the [nphies](#) that's used extensively in Saudi Arabia for health insurance transactions.



- <https://build.fhir.org/exchange-module.html>
- 1. Services: Allows invocation of complex services like decision support or workflow management, suitable for scenarios where simple CRUD actions are insufficient.
- 2. Database / Persistence: Uses FHIR resources directly within a database for data storage and retrieval, ensuring compatibility with FHIR-based APIs and other paradigms.
- 3. Subscription: Enables real-time notifications by allowing clients to subscribe to specific events or changes in FHIR resources, useful for timely updates like lab results or event detection.

- are comprehensive documents that combine multiple FHIR components to create a practical, real-world implementation for specific use cases.



- *“FHIR IGs = FHIR Profiles + Examples + Exchange Paradigm + LOTS of Documentation”*
- IGs serve as a blueprint for implementing FHIR in specific contexts, providing detailed guidance on how to use FHIR resources, profiles, and exchange paradigms to achieve particular healthcare interoperability goals.

- FHIR Implementation Guides usually include:
 - 1. Profiles and Extensions:** IGs define specific profiles and extensions tailored to the use case, constraining and extending FHIR resources as needed.
 - 2. Examples:** They provide concrete examples of how resources should be structured and used in the given context.
 - 3. Exchange Paradigms:** IGs specify which exchange methods (RESTful API, Documents, or Messaging) should be used and how.
 - 4. Detailed Documentation:** Extensive documentation explains how to implement the guide, including technical details, workflows, and best practices.

- US <https://build.fhir.org/ig/HL7/US-Core/index.html>
- ADBM <https://www.nrces.in/ndhm/fhir/r4/index.html>
- Nphies <https://de.scribd.com/document/719877031/NPHIES-Implementation-Guide-v2-0>
- Austria <https://fhir.hl7.at/HL7-AT-FHIR-Core-R5/index.html>
- **Registry:** <https://fhir.org/guides/registry/>
- <https://simplifier.net/>